

# Take-Home Assessment: Query Intelligence Endpoint

**Time:** 45–60 minutes | **AI tools allowed**

## Background

You're building the backend for an AI-powered research platform. Corporate users submit natural language queries like *"find battery technology startups in Southeast Asia"* and the system processes them to extract structured intelligence.

## Your Task

Given the starter code below (a bare FastAPI app and the Anthropic SDK installed), build a `/queries` feature with two endpoints:

### POST `/queries`

- Accept a natural language query string from the user
- Use an LLM to extract structured information from it
- Persist the query and the extracted data
- Return something useful to the caller

### GET `/queries/{id}`

- Return the stored query and its extracted data

That's the full spec. The rest is up to you.

## Starter Code

```
Python
# main.py
from fastapi import FastAPI
import anthropic
import os
```

```
app = FastAPI()
llm = anthropic.Anthropic(api_key=os.environ["ANTHROPIC_API_KEY"])
```

```
# Your code here
```

None

```
# .env
ANTHROPIC_API_KEY=sk-...
```

Python

```
# requirements.txt
fastapi
uvicorn
anthropic
python-dotenv
pydantic
```

Please include a short [README](#) (a few bullet points is fine) describing what you built and one thing you'd do differently with more time.